

1. (a) The DID model can be written as

$$score = \beta_0 + \beta_1 Treat^1 + \beta_2 Treat^2 + \beta_3 D^{NTU} + \beta_4 D^{2021} + \beta_5 D^{2022} + u,$$

where

$$Treat^1 = D^{NTU} \times D^{2021},$$

$$Treat^2 = D^{NTU} \times D^{2022},$$

$D^{NTU} = 1$ if the student is with NTU and 0 otherwise,

$D^{2021} = 1$ if the year is 2021 and 0 otherwise,

$D^{2022} = 1$ if the year is 2022 and 0 otherwise,

and u is the error.

The interpretation of the parameters is as follows:

$$E[score|D^{NTU} = 0, D^{2021} = 0, D^{2022} = 0] = \beta_0, \quad (1)$$

$$E[score|D^{NTU} = 0, D^{2021} = 1, D^{2022} = 0] = \beta_0 + \beta_4, \quad (2)$$

$$E[score|D^{NTU} = 0, D^{2021} = 0, D^{2022} = 1] = \beta_0 + \beta_5, \quad (3)$$

$$E[score|D^{NTU} = 1, D^{2021} = 0, D^{2022} = 0] = \beta_0 + \beta_3, \quad (4)$$

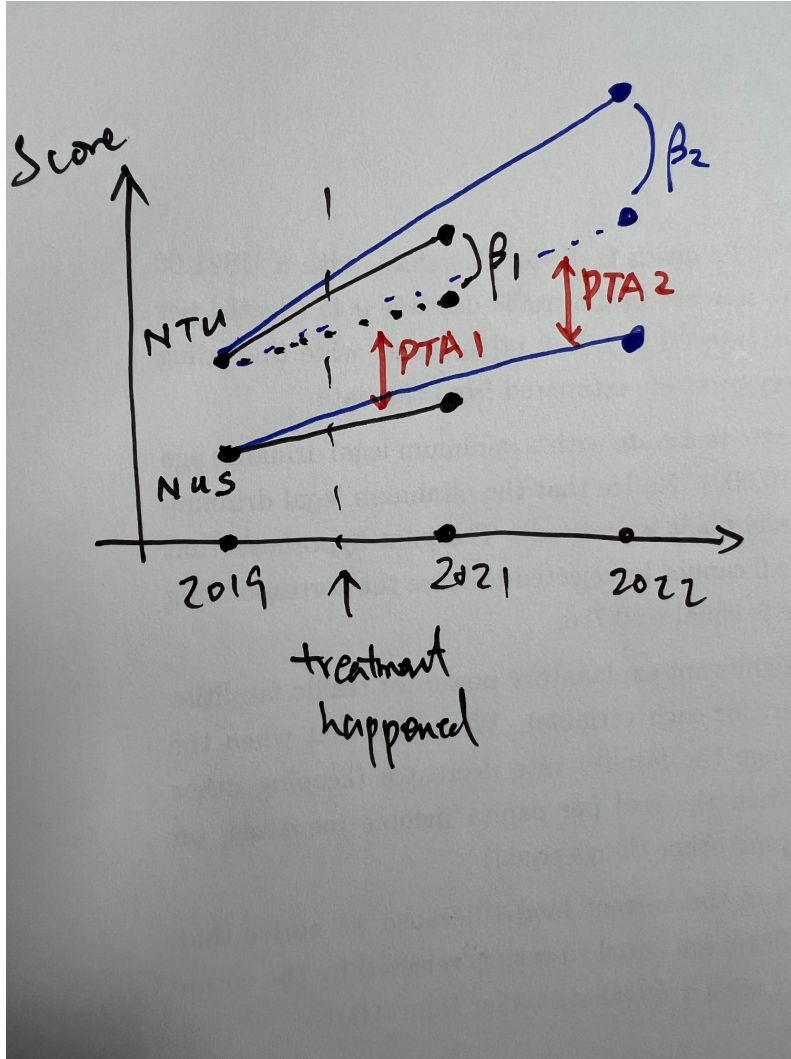
$$E[score|D^{NTU} = 1, D^{2021} = 1, D^{2022} = 0] = \beta_0 + \beta_1 + \beta_3 + \beta_4, \quad (5)$$

$$E[score|D^{NTU} = 1, D^{2021} = 0, D^{2022} = 1] = \beta_0 + \beta_2 + \beta_3 + \beta_5, \quad (6)$$

where (1) corresponds to control group (NUS), before the treatment happens (2019); (2) means control group, 1st period after the treatment happens (2021); (3) means control group, 2nd period after the treatment happens (2022); (4) means treatment group (NTU), before the treatment happens (2019); (5) means treatment group, 1st period after the treatment happens (2021); (6) means treatment group, 2nd period after the treatment happens (2022);

The DID effects for 2021 can be obtained by $[(5) - (4)] - [(2) - (1)]$, which is equal to β_1 . Similarly, that for 2022 can be obtained by $[(6) - (4)] - [(3) - (1)]$, which is equal to β_2 .

- (b) In the model, there are two parallel trend assumptions (PTA1 and PTA2 in the figure). Both assume that without the treatment, the NTU trend should move in the same way as the NUS trend. PTA1 is for the trend from 2019 to 2021, while PTA2 is for the trends from 2019 to 2022.



2. We note that the original model is equivalent to the following transformed model:

$$Y_i = \beta_0 + \gamma_1 X_{1i} + \beta_2 W_i + u_i, \quad (7)$$

where $\gamma_1 = \beta_1 - \beta_2$ and $W_i = X_{1i} + X_{2i}$.

Therefore, testing the original hypothesis $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$ is equivalent to testing $H_0^{transformed} : \gamma_1 = 0$ vs. $H_1^{transformed} : \gamma_1 \neq 0$.

Then, we can just run the OLS multiple regression by using Y_i as the dependent variable and (X_{1i}, W_i) as the explanatory variables. After getting the OLS estimator $\hat{\gamma}_1$ and its standard error $SE(\hat{\gamma}_1)$, we can test $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$ by testing $H_0^{transformed} : \gamma_1 = 0$ vs. $H_1^{transformed} : \gamma_1 \neq 0$ with the following t -test:

$$t = \frac{\hat{\gamma}_1 - 0}{SE(\hat{\gamma}_1)}. \quad (8)$$

We can reject $H_0 : \beta_1 = \beta_2$ at the 5% significance level if $|t| > 1.96$, etc.